Customer Data Analysis

Business Insights & Executive Summary

KPI	Value
Total Customers	2,100
Total Sales	1,021,324.45
Purchase Count	2,003
Average Purchase	509.90
Average Rating	3.03 / 5

Key Business Insights

1. Overall Sales: Total recorded sales reached **1,021,324.45** across 2,003 purchases, with an average purchase value of **509.90**.

2. Product Categories: Among categories with known product information, **Clothing** generated the highest sales at **163,864.98**, followed by Electronics at 154,691.51.

3. Missing Product Categories: A considerable number of records have an Unknown product category. These records were excluded from the category comparison so missing data is not treated as a real category.

4. Gender Distribution: The dataset contains **942 Male**, **891 Female**, and **267 Unknown** customers, showing relatively close Male and Female customer counts.

5. Customer Ratings: The average valid rating is **3.03/5**. Rating 3 has the highest number of valid ratings, with 326 observations.

6. Age: Among customers with valid age data, the average age is **54.37 years** and the median is **55 years**. However, about **76.43%** of Age values are missing or invalid, so age-based findings should be interpreted cautiously.

7. Sales Trend: The available data covers **October 2022 through July 2025**. 2024 has the highest total sales among the represented years, while 2022 and 2025 are partial periods.

Data Quality Notes

• 50 exact duplicate records were removed.

• Invalid ages below 15 or above 100 were treated as missing.

• Ratings outside the 1–5 range were treated as missing.

• Invalid purchase dates were converted to missing dates without removing the entire customer record.

• Missing Gender and ProductCategory values were labeled as Unknown.

• Missing PurchaseAmount values were not imputed, avoiding artificial changes to total sales.

Executive Summary

The analysis covers 2,100 customer records after duplicate removal. The dataset generated 1.02M in recorded sales with an average purchase of approximately 510. Clothing was the highest-selling known product category, while the average valid customer rating was 3.03/5. Gender counts were relatively balanced between Male and Female customers. The main data-quality limitation is the high percentage of missing or invalid Age values, followed by missing product-category information. These limitations should be considered when interpreting customer segmentation and category-level results.

Tools Used

Python was used for data cleaning, validation, and lightweight exploratory data analysis. Power BI was used for KPI calculations, visualizations, dashboard creation, and presentation of the final analysis.